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SciScore Report

Below you will find your SciScore report containing three tables. Your score is calculated based on adherence to scientific rigor criteria (Table 1) and identification of key biological resources (Table 2). Table 3 contains statistical tests and oligonucleotides but is not scored. If SciScore makes any mistakes, please contact us to help us learn and improve.

Table 1: Rigor Adherence Table

Ethics
IACUC: The animal studies were approved by the Animal Care and Ethics Committee of West China Hospital of Stomatology, Sichuan University.
Inclusion and Exclusion Criteria
not detected.
Attrition
not detected.
Sex as a biological variable
Male dbdb mice were used as animal model of T2D , and the heterozygous mice were used as control .
Subject Demographics
Age: Quantitative Reverse Transcription Polymerase Chain Reaction (qRT-PCR) A total of 8 dbdb mice and 8 heterozygous mice at the age of 8 weeks were used.
Randomization
not detected.
Blinding
not detected.
Power Analysis
not detected.
Replication
not required.
Data Information
Availability: 6 The data is publicly available on the DIAGRAM website (http:diagram-consortium.orgindex.html).

Identifiers: Table 1 The 20 Most Important m6A-SNPs Related to T2D Variant CHR Position P . value Gene Confidence_Level eQTL Hits DEG Methylation Type Allele Frequency rs1801206 4 6302707 4.90E-39 WFS1 Prediction: (Low) 8 hits No m6A 5.76E-01 rs116234738 5 102537357 1.40E-27 PPIP5K2 Prediction: (Low) 0 hit Yes m6A 3.07E-02 rs5213 11 17408404	rs1801206
	rs116234738
	rs5213
Identifiers: Medium) 16 hits Yes m6A 7.18E-01 rs750625 8 41525914	rs750625
Identifiers: ANK1 Prediction: (Low) 3 hits No m6A 6.57E-06 rs10832778 11 17394073 8.50E-23 NCR3LG1 Prediction: (Low) 26 hits No m6A 7.09E-01 rs9379084 6 7231843	rs10832778
	rs9379084
Identifiers: RREB1 MeRIP-Seq: (Medium) 1 hit Yes m6A 9.19E-02 rs12590 10 12292348 5.50E-17 CDC123 Prediction: (Low) 2 hits No m6A 1.50E-01 rs3743481 15 77907145 1.60E-16	rs12590
	rs3743481
Identifiers: LINGO1 MeRIP-Seq: (Medium) 1 hit No m6A 4.42E-01 rs2641348 1 120437884 5.60E-16 ADAM30 Prediction: (Low) 6 hits No m6A 1.69E-01 rs6685892 1 120458004 6.20E-16 NOTCH2 Prediction: (Low) 5 hits No m6A 1.68E-01 rs2793823 1 120437718 7.30E-15 ADAM30 Prediction: (Low) 3 hits No m6A 2.70E-01 rs11553326 4 724884 5.50E-12 PCGF3 Prediction: (Low) 7 hits No m6A 3.21E-02 rs2308891 6 32610009	rs2641348
	rs6685892
	rs2793823
	rs11553326
	rs2308891
Identifiers: HLA-DQA1 Prediction: (Low) 293 hits No m6A 2.44E-02 rs3760471 17 4069735 2.80E-11 ANKFY1 Prediction: (Low) 5 hits Yes m6A 1.38E-04 rs1058129 15 77905661 4.30E-11	rs3760471
	rs1058129
Identifiers: LINGO1 Prediction: (Low) 5 hits No m6A 7.06E-01 rs7250850 19 47585517 9.70E-11 ZC3H4 Prediction: (Low) 1 hit No m6A 5.23E-01 rs1046080 6 31595882 1.30E-10	rs7250850
	rs1046080
Identifiers: PRRC2A miCLIP: (High) 38 hits No m6A 7.43E-01 rs1033500 6 32307382 3.00E-10 C6orf10 Prediction: (Low) 21 hits No m6A 3.37E-01 rs2227956 6 31778272 3.10E-10	rs1033500
	rs2227956
Identifiers: HSPA1L Prediction: (Low) 112 hits No m6A 8.77E-01 rs4993986 6 32627652 6.10E-09 HLA-DQB1 miCLIP: (High) 1 hit Yes m6A 3.69E-01 Prediction of m6A Modification Near m6A-SNPs Among the T2D-associated m6A-SNPs identified above , we predicted the potential m6A modification site on the transcript by inputting the reference and altered sequence to an online m6A modification prediction tool , SRAMP (http:www.cuilab.cnsramp).	rs4993986
Identifiers: Three datasets GSE23343 , GSE38642 and GSE78721 containing standardized T2D-associated gene expression signals were retrieved and downloaded from GEO database (http:www.ncbi.nlm.nih.govgeo) .	GSE23343
	GSE38642
	GSE78721

Identifiers: Also , to prove that these gene expression differences were mediated by m6A-SNPs instead of the T2D phenotype , we downloaded the transcriptome data of tissues that are not closely related to the T2D phenotype as control sets , including enteroendocrine cells (GSE132831) and peripheral blood monocytes (GSE156061) .	GSE132831 GSE156061
Identifiers: For single-cell RNA sequencing (scRNA-seq) data analysis , we downloaded the islet scRNA-seq data of high-fat diet-induced diabetic mice and healthy controls in GSE162512 , and used the Seurat package to load the hdf5 format data for subsequent process.	GSE162512

Table 2: Key Resources Table

Your Sentences	REAGENT or RESOURCE	SOURCE	IDENTIFIER
Software and Algorithms			
6 The data is publicly available on the DIAGRAM website (http:diagram-consortium.orgindex.html).	DIAGRAM		Suggestion: diagram (RRID:SCR_026982)(link)
eQTL) effects through the HaploReg browser (http:pubs . broadinstitute.orgmammalshaploreghaploreg . php).32 Further , we also searched for several other potential regulatory mechanisms of these SNPs on transcriptional regulation , such as altering protein binding and changing motif .	HaploReg		Suggestion: HaploReg (RRID:SCR_006796)(link)
For single-cell RNA sequencing (scRNA-seq) data analysis , we downloaded the islet scRNA-seq data of high-fat diet-induced diabetic mice and healthy controls in GSE162512 , and used the Seurat package to load the hdf5 format data for subsequent process.	Seurat		

Other Entities Detected

Your Sentences	Recognized Entity
Statistical Tests	
Considering that multiple tests are used among the three datasets , we use the Bonferroni adjustment to set a more stringent threshold , $p < 0.0167$ (0.053) was set as a threshold for defining differential genes .	Bonferroni

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For a full description of scored criteria and tips for improving your score, please see https://www.scicrunch.com/sciscorereport-faq